

Finite-Sample Behaviour of Percentile Bootstrap Confidence Intervals for Bernoulli Proportions

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Abstract

Percentile bootstrap confidence intervals are widely used due to their simplicity and general applicability. However, their performance can deteriorate when the underlying estimator is discrete, as in the case of the sample proportion in a Bernoulli model. This paper examines the finite-sample behavior of percentile bootstrap intervals using analytical tools that characterize bootstrap order statistics through binomial probabilities. We specialize these results to the Bernoulli proportion and assess the practical implications through simulation. The findings highlight structural coverage deficiencies inherent to the percentile method in discrete settings.

1 Introduction

Bootstrap methods have become a flexible approach to quantify uncertainty and are well accepted in almost all fields of statistics. In particular, the bootstrap percentile method is often most attractive because of its simplicity and low modeling effort. However, the method can be quite erratic in those situations when the sampling distribution of the estimator is non smooth, particularly so in situations where the estimators are constrained to a closed and/or very small set. A clear example is the case of the Bernoulli proportion.

Here, our focus is to understand how the percentile bootstrap method behaves in case of a discrete estimator using the Bernoulli proportion as a case in point. We combine the most recent theoretical work and simulation to explain the reason for the method to have very unequal and/or biased coverage.

2 Theoretical Background

This section summarises the analytical foundations needed to understand the finite-sample behavior of percentile bootstrap confidence intervals. Following the approach of (1), the key

observation is that bootstrap order statistics can be characterised exactly through binomial tail probabilities. This representation is especially useful for analysing discrete estimators.

2.1 Percentile Bootstrap Interval

Let $X_1, \dots, X_n$ be independent and identically distributed (i.i.d.) from a distribution F with parameter θ , and let $\hat{\theta} = T(X_1^n)$ be an estimator. Generate m bootstrap replicates $\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(m)}$ either by parametric resampling from $F_{\hat{\theta}}$ or by nonparametric resampling with replacement from the observed data.

Let $u_{(1)} \leq \dots \leq u_{(m)}$ denote the ordered bootstrap replicates. For a nominal level $1 - \alpha$, define

$$l = \left\lfloor \frac{m\alpha}{2} \right\rfloor + 1, \quad u = m + 1 - l.$$

The **percentile bootstrap interval** is then

$$I_{\text{perc}} = [u_{(l)}, u_{(u)}]. \quad (1)$$

2.2 Distribution of Bootstrap Order Statistics

Let

$$H(x) = P(\hat{\theta}^* \leq x \mid X_1^n)$$

be the conditional bootstrap CDF. Since the bootstrap replicates are conditionally i.i.d., the order statistics of the bootstrap sample satisfy

$$P(u_{(j)} \leq x \mid X_1^n) = 1 - F_B(j - 1; m, H(x)), \quad (2)$$

where $F_B(\cdot; m, p)$ is the binomial CDF. This representation is central, which means that it expresses the bootstrap order statistics entirely in terms of H and binomial probabilities, enabling exact finite-sample analysis.

2.3 Finite-Sample Coverage

The percentile interval contains θ when

$$u_{(l)} \leq \theta \leq u_{(u)}.$$

Using the order-statistic identity above, (1) express the exact coverage of the parametric bootstrap as

$$\text{Cover}_{pa}(\theta) = E_{\hat{\theta}} \left[F_B(u - 1; m, H_{\hat{\theta}}(\theta^-)) - F_B(l - 1; m, H_{\hat{\theta}}(\theta)) \right],$$

with an analogous expression for the nonparametric bootstrap.

Although we do not evaluate these formulas directly in our simulation study, they emphasise that coverage depends on the smoothness of the bootstrap distribution function $H(x)$.

For continuous estimators, $H(x)$ is typically smooth, and coverage tends to be stable.

For discrete estimators, however, $H(x)$ becomes a step function, and the binomial-tail representation amplifies the discontinuities, leading to oscillatory or biased coverage, a phenomenon illustrated clearly in the Bernoulli model discussed next.

3 Percentile Bootstrap for the Bernoulli Proportion

The Bernoulli model illustrates how the percentile bootstrap inherits the discrete structure of the estimator, producing irregular quantile behaviour and erratic coverage.

3.1 Bootstrap Distribution for the Bernoulli Case

Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$ and $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$. Conditional on $\hat{p} = k/n$, parametric and nonparametric bootstrap samples both generate

$$\hat{p}^* \mid \hat{p} = \frac{k}{n} = \frac{1}{n} \cdot \text{Binomial}\left(n, \frac{k}{n}\right).$$

Thus, the bootstrap distribution is supported on the grid $\{0, 1/n, \dots, 1\}$ and is a discrete step function whose jumps depend on k . The percentile bootstrap interval therefore inherits this discrete structure.

3.2 Coverage Implications

Let $H_k(x) = P(\hat{p}^* \leq x \mid \hat{p} = k/n)$ denote the bootstrap distribution. Using the general coverage identity, the coverage probability at the parameter value p can be written as

$$P_p(p \in I_{\text{perc}}) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} \mathbf{1}\left\{p \in [Q_{\alpha/2}(H_k), Q_{1-\alpha/2}(H_k)]\right\}.$$

Although we do not compute this expression numerically, it reveals that the coverage changes only when the bootstrap quantiles change, and these quantiles change only at the jump points of H_k . This produces oscillatory, non-smooth coverage behaviour.

3.3 Structural Limitations of the Percentile Method

Three mechanisms drive the irregular coverage observed in the Bernoulli case:

1. **Quantile monotonicity failure:** because $H_k(x)$ is discrete, its quantiles do not vary smoothly with k/n .
2. **Boundary degeneracy:** when $\hat{p} = 0$ or $\hat{p} = 1$, the bootstrap distribution collapses to a point mass, resulting in zero-length intervals and severe undercoverage near the boundaries.
3. **Persistence under a large bootstrap size:** increasing m refines the empirical bootstrap distribution, but does not eliminate its discrete jumps.

4 Simulation Study

We consider

$$p \in \{0.01, 0.05, 0.1, 0.3, 0.5, 0.8, 0.99\}, \quad n \in \{10, 30, 100\}, \quad m \in \{100, 1000, 5000\},$$

with $M = 2000$ Monte Carlo repetitions. We compare the percentile bootstrap (nonparametric and parametric) with Wilson and Clopper–Pearson intervals.

4.1 Coverage

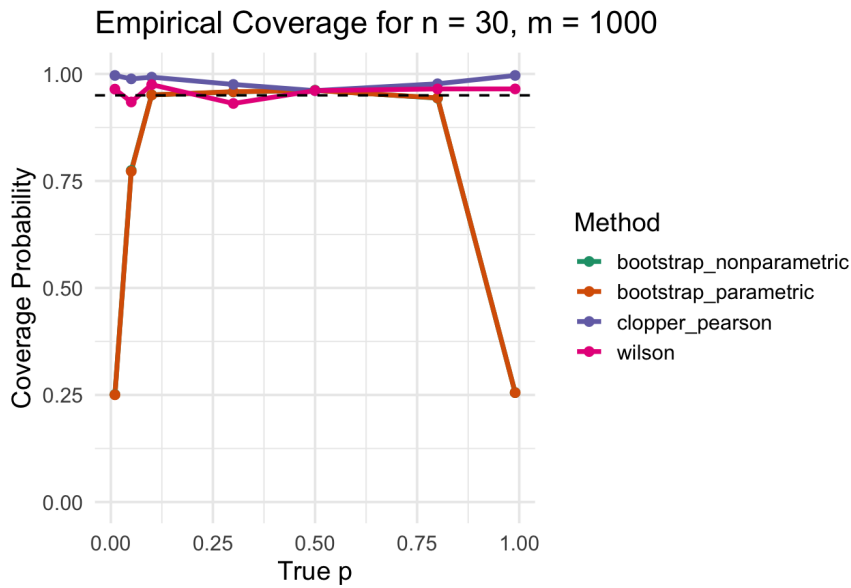


Figure 1: Empirical coverage for $n = 30, m = 1000$.

The percentile interval shows severe undercoverage at $p = 0.01$ and $p = 0.99$, dropping to values below 75% in some cases. In intermediate regions (e.g. $p = 0.3$ or 0.5), the coverage oscillates due to the stepwise nature of the bootstrap quantiles: as p increases smoothly, the bootstrap distribution only changes when $\hat{p}$ moves to a different lattice point. The parametric bootstrap performs slightly better but still displays noticeable irregularity. By contrast, the Wilson interval remains close to the nominal 95% level throughout the full parameter range.

4.2 Expected Length

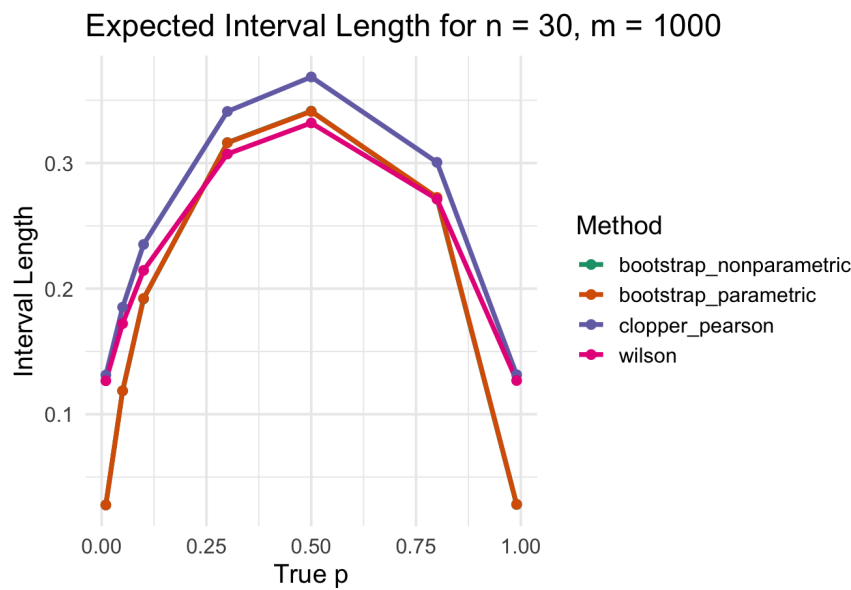


Figure 2: Expected interval length for $n = 30$.

The percentile bootstrap intervals are consistently shorter across all values of p . Although this might suggest higher efficiency, the shorter average length primarily reflects a systematic underestimation of uncertainty.

By contrast, Wilson intervals achieve competitive lengths while maintaining accurate coverage, indicating a more favourable balance between bias and variance.

4.3 Interval Length Variability

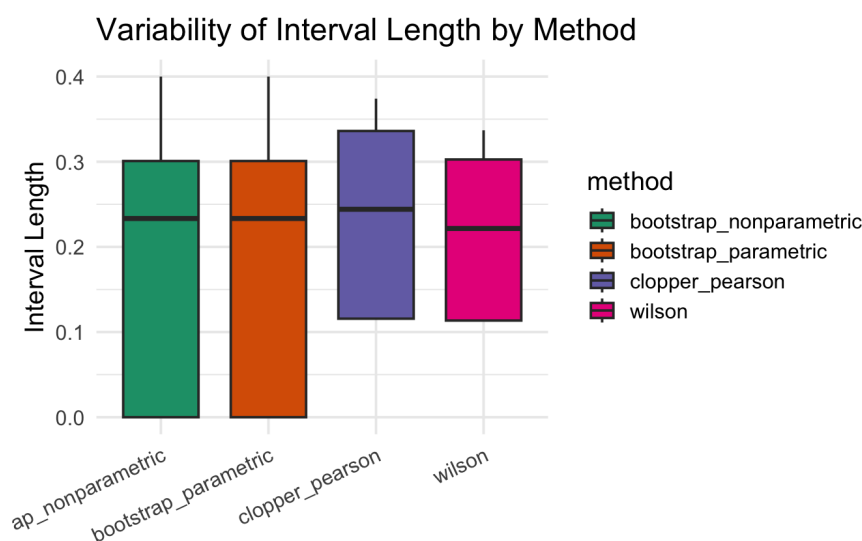


Figure 3: Distribution of interval lengths across methods.

The boxplot reveals clear differences in interval-length stability across methods. Percentile bootstrap intervals show the greatest variability due to the discreteness of the Bernoulli estimator: whenever $\hat{p}$ takes the values 0 or 1, the bootstrap distribution collapses to a point mass, generating intervals of length zero.

Wilson intervals exhibit markedly lower variability, as their analytic construction prevents interval collapse and stabilises interval width even in extreme samples.

Clopper–Pearson intervals are also stable, though for a different reason: their conservative nature yields consistently wider intervals, reducing variation across repetitions.

Overall, these results confirm that bootstrap-based intervals are intrinsically unstable in discrete settings, whereas Wilson and Clopper–Pearson intervals gain stability from structural properties unrelated to resampling.

5 Conclusion

The behaviour of percentile-bootstrap confidence intervals was studied by merging recent analytical results to develop a simulation study pertaining to Bernoulli proportions. The analysis had a theoretical component, from which several inherent weaknesses of the percentile bootstrap approach were identified. In particular, the empirical bootstrap distribution is stepwise, its quantiles change only at discrete jump points, and the distribution becomes degenerate whenever the empirical proportion equals 0 or 1. These mechanisms jointly explain the predicted coverage gaps and the instability observed in bootstrap-based confidence intervals.

The simulation results confirmed these theoretical predictions. Both the parametric and nonparametric percentile-bootstrap intervals systematically undercover, especially near the boundaries, and display irregular behaviour in intermediate regions. Their intervals tend to be shorter on average, but this does not reflect improved efficiency; rather, it indicates an underestimation of uncertainty.

By contrast, Wilson intervals achieved coverage close to the nominal level and maintained stable lengths across all scenarios. Clopper–Pearson intervals, although conservative, also produced highly consistent interval widths. These properties make both methods substantially more reliable than the percentile bootstrap in finite-sample discrete settings.

Overall, the results show that the percentile bootstrap is unreliable for Bernoulli proportions in moderate sample sizes, and Wilson-like approaches should be preferred.

References

- [1] Wang, W., Yu, C., & Zhang, Z. (2025). *Finite-sample analytic properties of percentile bootstrap intervals*. *Metrika*.